

Auteur: Dara van Engelen
Studierichting: Informatica
Oefeningenreeks 4A nr. 37.11

$$\begin{aligned}\int \frac{dx}{4x^2 + 12x + 18} &= \int \frac{dx}{(2x+3)^2 + 9} \\ \text{Stel } 2x+3 &= t \Rightarrow 2dx = dt \Rightarrow dx = \frac{dt}{2} \\ &= \frac{1}{2} \int \frac{dt}{t^2 + 9} = \frac{1}{2} \int \frac{dt}{t^2 + 3^2} \\ &= \frac{1}{2} \cdot \frac{1}{3} \operatorname{Bgtan} \left(\frac{t}{3} \right) + C = \frac{1}{6} \operatorname{Bgtan} \left(\frac{2x+3}{3} \right) + C\end{aligned}$$

References